

Tables and columns for the following

Case 1: A Simple Case

- A video store rents movies to members.
- Each movie in the store has a title and is identified by a unique movie number.
- A movie can be in VHS, VCD, or DVD format.
- Each movie belongs to one of a given set of categories (action, adventure, comedy, ...)
- The store has a name and a (unique) phone number for each member.
- Each member may provide a favorite movie category (used for marketing purposes).
- There are two types of members:
 - Golden Members:
 - Bronze Members:
- Using their credit cards gold members can rent one or more movies and bronze members max. of one movie.
- A member may have a number of dependents (with known names).
- Each dependent is allowed to rent one (1) movie at a time.

Infos:

Movie-related

- Each movie has:
 - movie_id (unique)
 - title
 - format → VHS / VCD / DVD
 - belongs to one category

Category

- Fixed set: action, comedy, etc.
- One category → many movies

Member

- Each member has:
 - member_id
 - name
 - phone (unique)
 - type → GOLDEN / BRONZE
 - optional favorite category

Golden Member

- Has:
 - credit card
- Can rent multiple movies

Bronze Member

- No extra attributes
- Can rent max 1 movie at a time

Dependent

- Belongs to a member
- Has only:
 - name
- Can rent only 1 movie at a time

Rental

- A movie is rented by:
 - either a member OR a dependent
 - Has:
 - rental_date
 - return_date
-

CREATE TABLE QUERIES :

-- Table 1

```
CREATE TABLE Category (  
    category_id INT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,  
    name VARCHAR(50) NOT NULL UNIQUE  
);
```

-- Table 2

```
CREATE TABLE Movie (  
    movie_id INT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,  
    title VARCHAR(100) NOT NULL,  
    format VARCHAR(20) NOT NULL CHECK (format IN ('VHS', 'VCD', 'DVD')),  
    category_id INT NOT NULL,  
    FOREIGN KEY (category_id) REFERENCES Category(category_id)  
);
```

Table 3:

```
CREATE TABLE Member (  
    member_id INT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,  
    name VARCHAR(100) NOT NULL,  
    phone VARCHAR(10) NOT NULL UNIQUE CHECK (phone ~ '^d{10}$'),  
    member_type VARCHAR(10) NOT NULL CHECK (member_type IN ('GOLDEN', 'BRONZE')),  
    fav_category_id INT DEFAULT NULL,                -- optional  
    credit_card_number VARCHAR(20) DEFAULT NULL,  
    FOREIGN KEY (fav_category_id) REFERENCES Category(category_id)  
);
```

-- Table 4

```
CREATE TABLE Dependent (  
    dependent_id INT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,  
    member_id INT NOT NULL,  
    name VARCHAR(100) NOT NULL,  
    FOREIGN KEY (member_id) REFERENCES Member(member_id)  
        ON DELETE CASCADE  
);
```

-- Table 5

```
CREATE TABLE Rental (  
    rental_id INT GENERATED ALWAYS AS IDENTITY PRIMARY KEY,  
    movie_id INT NOT NULL,  
    rented_by_member_id INT DEFAULT NULL,  
    rented_by_dependent_id INT DEFAULT NULL,  
    rental_date DATE NOT NULL,  
    return_date DATE DEFAULT NULL,  
  
    FOREIGN KEY (movie_id) REFERENCES Movie(movie_id),  
    FOREIGN KEY (rented_by_member_id) REFERENCES Member(member_id),  
    FOREIGN KEY (rented_by_dependent_id) REFERENCES Dependent(dependent_id),
```

--optional

```
    CONSTRAINT chk_one_renter CHECK (  
        (rented_by_member_id IS NOT NULL) !=  
        (rented_by_dependent_id IS NOT NULL)  
    )  
);
```

Optional Improvements:

1.If strict rule for GOLDEN CARD MEMBERS:

```
CONSTRAINT chk_card_rule CHECK (  
    (member_type = 'GOLDEN' AND credit_card_number IS NOT NULL) OR  
    (member_type = 'BRONZE')  
)
```

2. Dependent -> max 1 movie at a time

```
CREATE UNIQUE INDEX unique_active_dependent  
ON Rental(rented_by_dependent_id)  
WHERE return_date IS NULL;
```